

Multiple Choice (10 marks)

1. How many joules are equivalent to one lit-atmof work?
 - (a) 106.4 joule
 - (b) 98.6 joule
 - (c) 99.5 joule
 - (d) 97.8 joule
 - (e) 101.3 joule
2. Determine if the function $f(x, y) = [(x^2 - y^2) / (x^2 + y^2)](1)$ is discontinuous at the origin.
 - (a) The function is continuous at (0, 0)
 - (b) The function is continuous along the line $y = x$
 - (c) The function has a removable discontinuity at the origin
 - (d) The function is continuous along the line $y = -x$
 - (e) The function is continuous at all points
3. In force-current analogy, electrical analogous quantity for displacement (x) is
 - (a) inductance.
 - (b) resistance.
 - (c) impedance.
 - (d) voltage.
 - (e) capacitance.
4. Find the Laplace transform of $f(t) = t$, for $t > 0$.
 - (a) $L\{t\} = (2 / s^3)$
 - (b) $L\{t\} = (1 / s^3)$
 - (c) $L\{t\} = s^2$
 - (d) $L\{t\} = (1 / (2 s^2))$
 - (e) $L\{t\} = (1 / s^2)$
5. Use the derivative property of Laplace transforms to solve the differential equation $y' - y = e^{-x}(1)$ where $y(0) = 0$ is the initial value of y .
6. $\sinh x$

7. $e^x - e^{-x}$

7. $e^x + x$

8. $x^2 e^x$

9. $\cosh x$

True / False (10 marks)

Answer True or False

6. True or False: The convolution of $f(t) = t$ and $g(t) = t^2$ results in $f(t) * g(t) = t^4 / 12$.
7. True or False: The voltage across an inductor is equal to the current multiplied by the inductance, resulting in $V_L(t) = 0.1 \sin(10^6 t)$.
8. True or False: The gain of a half-wave dipole antenna is approximately 1.2 dBi.
9. True or False: A heat sink is sufficient on its own to protect the SCR from overcurrent without additional protective devices.
10. True or False: A steel bar designed for the same load and safety factor could adequately perform with a diameter of 2 inches.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between droplet diameter and internal pressure in water at 20°C. What diameter is required for the internal pressure to exceed external pressure by 1.0 kPa?
11. Using dimensional analysis, derive and discuss the relationship between the bubble diameter and the physical properties of water in the context of gas bubbles formed from a small orifice.

End of Paper